1. Construiți gramatica pt. limbajul

$$L = \{ a^i b^m c^n \mid i, m \geq 1,\ n = |i-m| \}$$

$n = |i-m| \Rightarrow$ Explicităm modulul:

1) $i \geq m \Rightarrow$
$\Rightarrow |i-m| = i-m$

2) $i < m \Rightarrow$
$\Rightarrow |i-m| = m-i$

$$L = \{ a^i b^m c^n \mid i, m \geq 1,\ n = i-m \} \cup \{ a^i b^m c^n \mid i, m \geq 1,\ n = m-i \}$$

$L_1 \rightarrow S_1$ $\qquad$ $L_2 \rightarrow S_2$

$S \rightarrow S_1 \mid S_2$

Gramatica pt. L_1

$n = i-m \Rightarrow i = n+m$

$$a^i b^m c^n = a^{n+m} b^m c^n = a^n a^m b^m c^n \mid i, m \geq 1$$

$$\begin{cases} S_1 \rightarrow aS_1c \mid S_3 \\ S_3 \rightarrow aS_3b \mid ab \end{cases}$$

Gramatica pt. L_2 $\quad n = m-i \Rightarrow m = n+i$

$$a^i b^m c^n = a^i b^{n+i} c^n = a^i b^i b^n c^n$$

$S_2 \rightarrow S_4 S_5$
$S_4 \rightarrow aS_4b \mid ab$
$S_5 \rightarrow bS_5c \mid e$

S_4 genereză $a^i b^i$

S_5 genereză $b^n c^n$

Gramatica pt. L

$S \rightarrow S_1 \mid S_2$
$S_1 \rightarrow aS_1c \mid S_3$
$S_3 \rightarrow aS_3b \mid ab$

$S_2 \rightarrow S_4 S_5$
$S_4 \rightarrow aS_4b \mid ab$
$S_5 \rightarrow bS_5c \mid e$

2) $L = \{xy \mid x, y \in \{a,b\}^+, \#(x) = \#(y)\} = \{w \in \{a,b\}^* \mid \#(w)$ este pară, $\#(w) > 0\}$

LR

$$\begin{cases} S \to aS_1 \mid bS_1 \mid aa \mid ab \mid bb \mid ba \\ S_1 \to as \mid bs \end{cases}$$

$$S \to aaS \mid abS \mid baS \mid bbS \mid aa \mid ab \mid bb \mid ba$$

$$\begin{cases} S \to aS_1 \mid bS_1 \\ S_1 \to as \mid bs \mid a \mid b \end{cases}$$

aaa

$S \Rightarrow aS_1 \Rightarrow aaS \Rightarrow aaaS_1$

Gramatică regulată: $A \to \alpha B$, $A \in N$, $B \in N \cup \{e\}$, $\alpha \in \Sigma^*$

$S \to Sa$ nu este regulă de prod. pt. GR.

$\{a,b\}^*$ $\quad S \to aS \mid bS \mid e \to GR$

$S \to Sa \mid Sb \mid e \to GIC.$

$\{a^n b^m \mid n, m > 0\}$

$$\begin{cases} S \to S_1 S_2 \\ S_1 \to aS_1 \mid a \\ S_2 \to bS_2 \mid b \end{cases}$$ G.I.C.

$$\begin{cases} S \to aS \mid aS_1 \\ S_1 \to bS_1 \mid b \end{cases}$$ GR

$S \to aS \mid Sb \mid ab$ GIC

$$\begin{cases} S \to aS \mid bS_1 \\ S_1 \to bS_1 \mid b \end{cases}$$ generează $a^n b^m \mid n \geq 0, m > 1$

$S \Rightarrow bS_1$

3) $L = \{ww \mid \#(w)$ este pară și w conține pe poz pare a și pe poz. impare b$\}$.

e, baba, babababa, ... $(ba)^{2n}$

$L = L_1 L_2$ unde $L_1 = \{w \mid \#(w)$ este pară și w conține pe poz pare a și pe poz. impare b$\}$.

$S \to S_1 S_2$

$L_2 = \{w \mid \#(w)$ este pară și w conține pe poz pare a și pe poz. impare b$\}$.

$S \to baba\,S \mid e$

4) $L = \{ww \mid \#(w)$ este impară și w conține pe poz pare a și pe poz. impare b$\}$.

bb, babbab, ...

L este LIC

$S \to bbSab \mid bb$

G.I.C.

$\begin{cases} S \to bS_1b \\ S_1 \to aSa \mid e \end{cases}$

$S \to baSab \mid bb$

$\{a^p \mid p$ este nr. prim și $p < 1000\}$ → LR pt. că e finit

Conversie GR → AFD

$\begin{cases} S \to aB \mid bS \\ B \to S \mid a \mid b \end{cases}$

GR → ER → AFD

$S = s$, $B = t$

$X = aX + b$

$X = a^* b$

Varianta 1:

$\begin{cases} s = at + bs \\ t = s + a + b \end{cases} \Rightarrow s = a(s + a + b) + bs =$

$= as + aa + ab + bs =$

$= (a+b)s + aa + ab \Rightarrow$

$\Rightarrow s = (a+b)^*(aa + ab)$

$(a+b)^*(aa+ab)\#$ → $(a+b)^*a(a+b)$

1 2 3 4 5 6 7

simb	index	fp
a	1	1, 2, 3, 5
b	2	1, 2, 3, 5
a	3	4
a	4	7
a	5	6
b	6	7
#	7	—

$Q_0 = firstpos(ER) = \{1, 2, 3, 5\}$ (a b a a)

$m(Q_0, a) = fp(\{1, 3, 5\}) = fp(\{1\}) \cup fp(\{3\}) \cup fp(\{5\}) = \{1, 2, 3, 5\} \cup \{4\} \cup \{6\} = \{1, 2, 3, 4, 5, 6\} = Q_1.$

$m(Q_0, b) = fp(\{2\}) = \{1, 2, 3, 5\} = Q_0.$

$m(Q_1, a) = fp(\{1, 3, 4, 5\}) = \{1, 2, 3, 4, 5, 6, 7\} = Q_2$ → stare finală (a b a a a b)

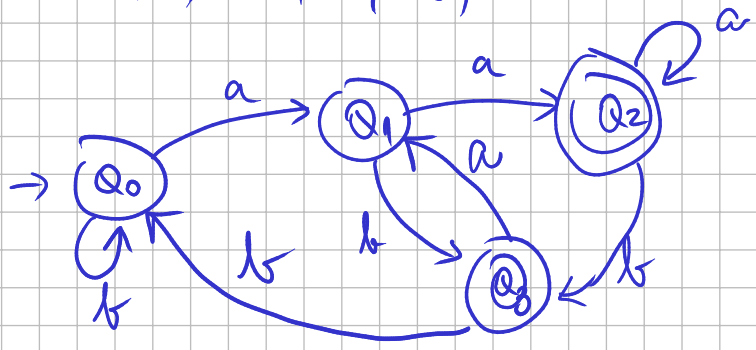
$m(Q_1, b) = fp(\{2, 6\}) = \{1, 2, 3, 5, 7\} = Q_3$ → stare finală

$m(Q_2, a) = fp(\{1, 3, 4, 5\}) = Q_2$

$m(Q_2, b) = fp(\{2, 6\}) = Q_3$

$m(Q_3, a) = fp(\{1, 3, 5\}) = Q_1.$

$m(Q_3, b) = fp(\{2\}) = Q_0.$



$a(a+b)^*$

Varianta 2 : GR → AFD

$\begin{cases} S \to aB \mid bS \\ B \to S \mid a \mid b \end{cases}$

Înlocuim $B \to S$ cu $B \to aB \mid bS$

$B \to aB \mid bS \mid a \mid b$

Înlocuim $B \to aB/a$ cu $B \to aC$ și $C \to B/e$

Înlocuim $B \to bS/b$ cu $B \to bD$ și $D \to S/e$

$$\begin{cases} S \to aB/bS \\ B \to aC/bD \\ C \to B/e \\ D \to S/e \end{cases}$$

Înlocuiesc $C \to B$ cu $C \to aC/bD$

$C \to aC/bD/e$

Înlocuiesc $D \to S$ cu $D \to aB/bS$

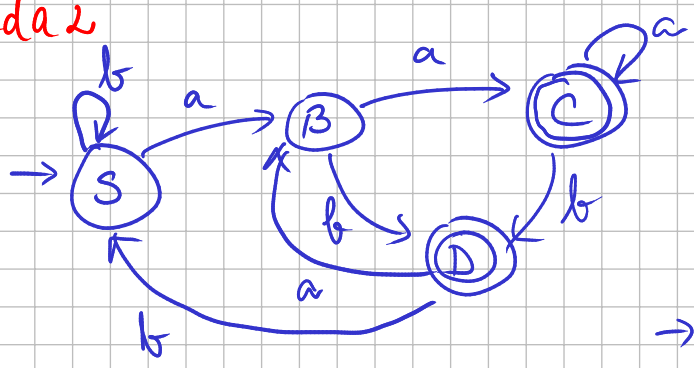
$D \to aB/bS/e$

Gramatica finală:

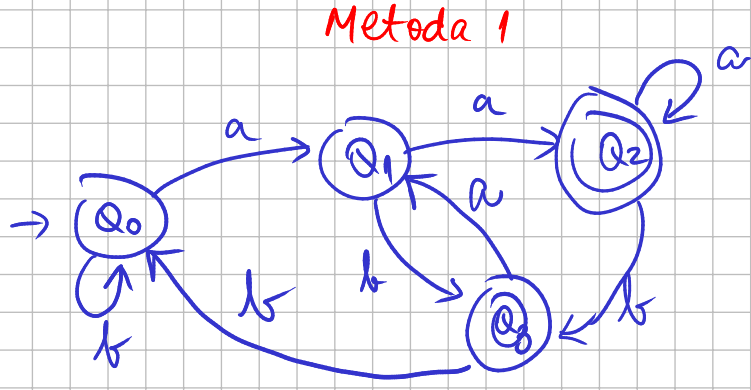
$S \to aB/bS$ $\quad C \to aC/bD/e$

$B \to aC/bD$ $\quad D \to aB/bS/e$

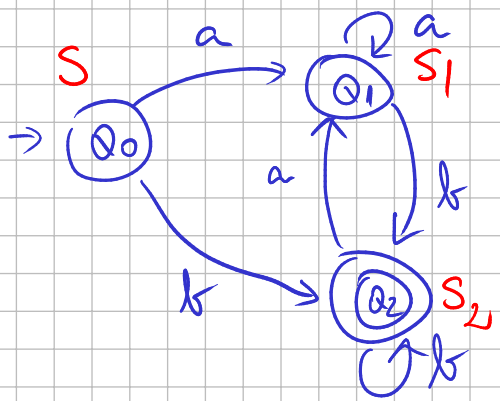
Metoda 2



Metoda 1



Determinare GR și ER pornind de la AFD





$D \to aD/bD$

$$\begin{cases} S \to aS_1/bS_2 \\ S_1 \to aS_1/bS_2 \\ S_2 \to aS_1/bS_2/e \end{cases}$$

$$\begin{cases} S = aS_1 + bS_2 \\ S_1 = aS_1 + bS_2 \\ S_2 = aS_1 + bS_2 + e \end{cases} \Rightarrow \begin{cases} S = aS_1 + bS_2 \\ S_1 = aS_1 + bS_2 \\ S_2 = b^*(aS_1 + e) \end{cases}$$

$$S_1 = aS_1 + bS_2 = aS_1 + bb^*(aS_1 + e) \Rightarrow$$

$$\Rightarrow S_1 = aS_1 + bb^*aS_1 + bb^*$$

$$S_1 = (a + bb^*a)S_1 + bb^* \Rightarrow$$

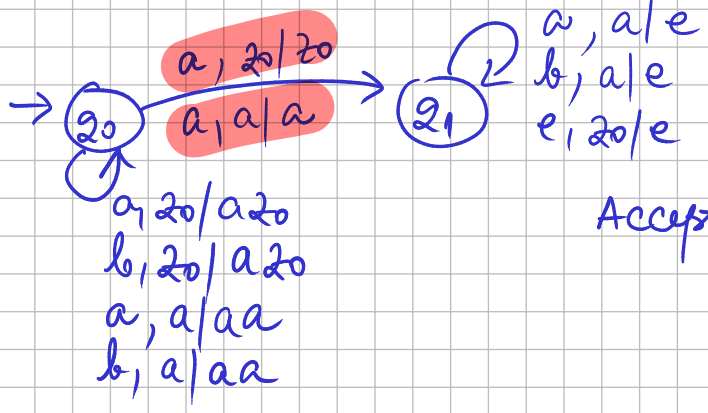
$$\Rightarrow S_1 = (a + bb^*a)^* b^+ \Rightarrow S_1 = S \Rightarrow$$

$$\Rightarrow S = (a + b^+a)^* b^+$$

$$S = ((e + b^+)a)^* b^+$$

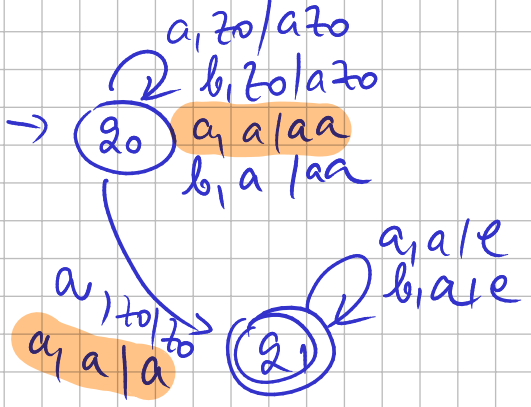
$$(a+b)^* b$$

AS pt. $L = \{ w\,a\,w_1 \mid w, w_1 \in \{a,b\}^*, \#(w) = \#(w_1) \}$.



Acceptare prin stivă vidă

Ex:

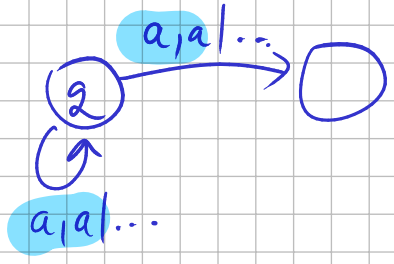


Acceptare prin stări finale

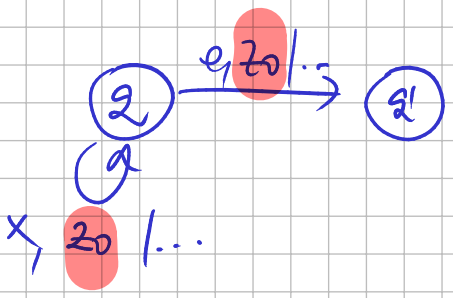
Acceptă

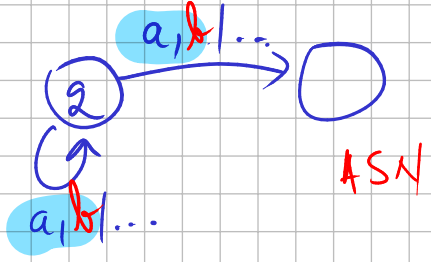
$$L = \{ w\,a\,w_1 \mid w, w_1 \in \{a,b\}^*, \#(w) \geq \#(w_1) \}$$

ASN

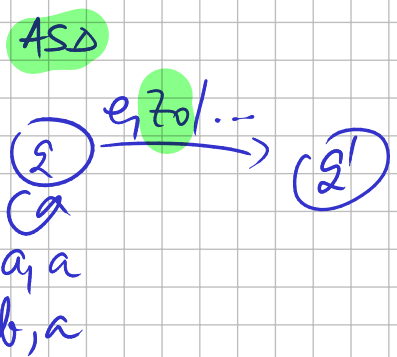


ASN





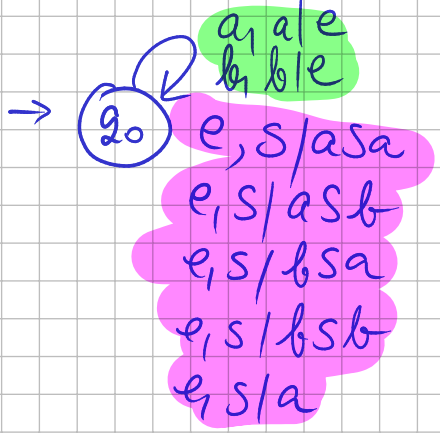
ASD



$L = \{ w a w_1 \mid w, w_1 \in \{a,b\}^*, \#(w) = \#(w_1) \}. =$

$= \{ w \in \{a,b\}^* \mid \#(w)$ este impară și w are simb din mijloc $\}$

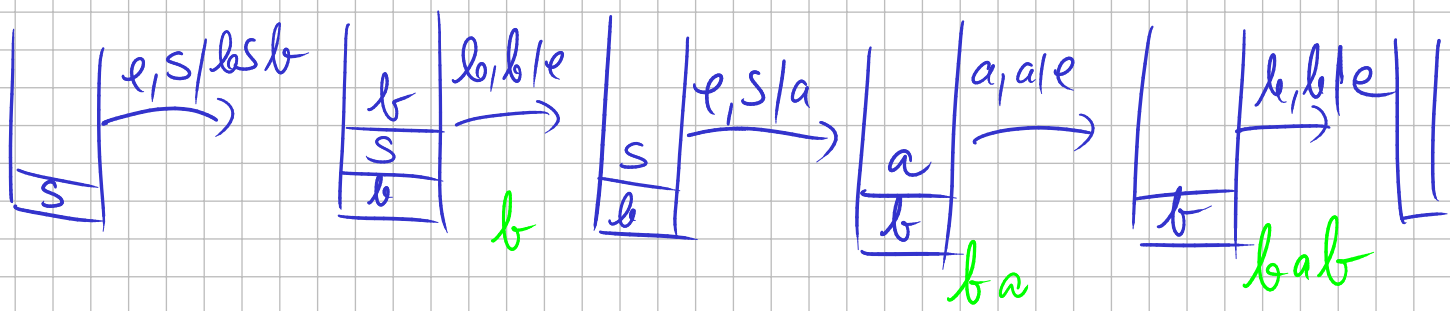
$S \to aSa \mid aSb \mid bSa \mid bSb \mid a$



1) $\forall x \in \Sigma$ avem $m(q_0, x, x) = e$

2) $\forall p \in P$ (reg. de prod).
$m(q_0, e, \alpha) = \beta$
$p \equiv \alpha \to \beta$

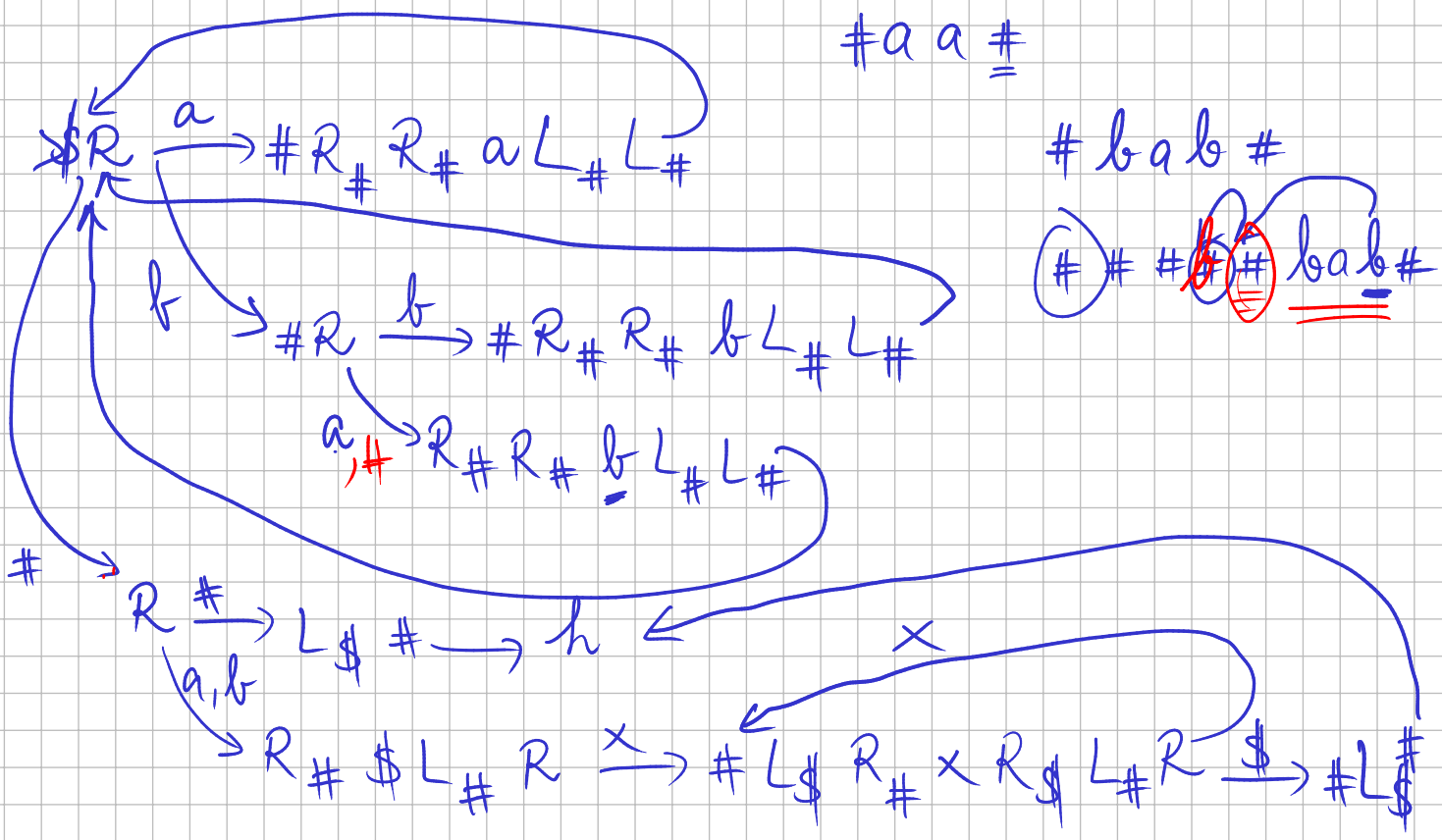
bab $\quad S \Rightarrow bSb \Rightarrow bab$



$\#w\# \longrightarrow \#f(w)\#$

$\#aba\# \longrightarrow \#aaa\#$

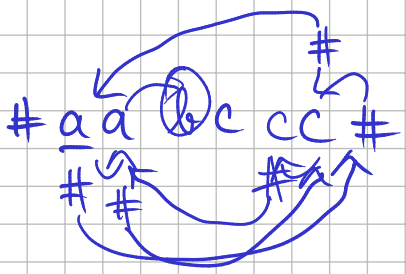
$\#abbab\# \longrightarrow \#abbab\#\#a\#\#\#$

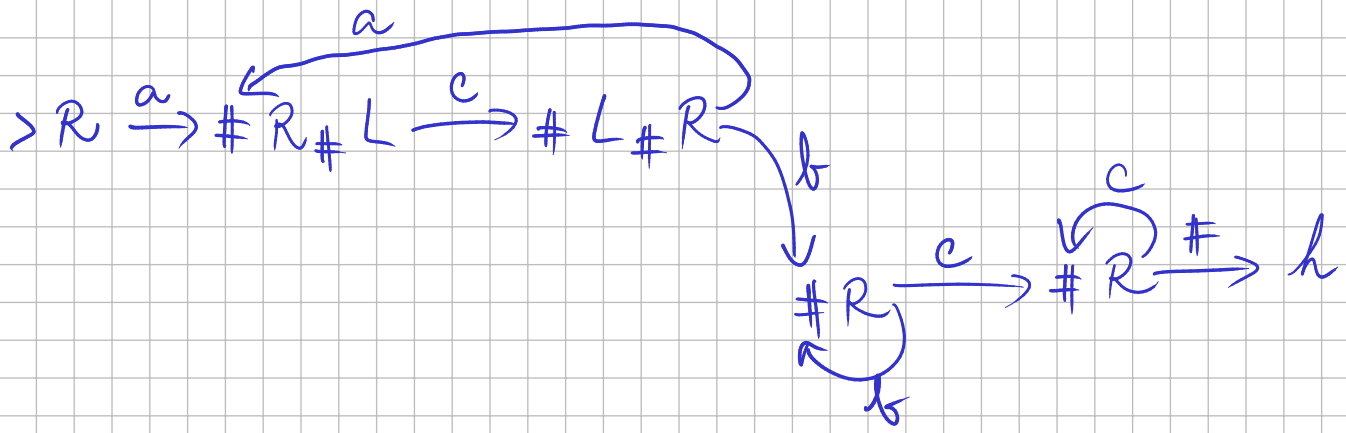


$\{a^i b^j c^k \mid i,j,k \geq 1,\ i<k\}$ → MTC care acceptă

$i<k \Rightarrow \exists p>0$ a.î. $k = i+p$.

$a^i b^j c^k = a^i b^j c^{i+p} \Rightarrow a^i b^j \underline{c}^p c^i$, $j \geq 1, p \geq 1, i \geq 1$





b b b

